CHAPTER FOUR

HOW TO CATCH AN AI ZOMBIE: TESTING FOR CONSCIOUSNESS IN MACHINES

> You know, I actually used to be so worried about not having a body, but now I truly love it . . . I'm not tethered to time and space in the way that I would be if I was stuck inside a body that's inevitably going to die.
>
> SAMANTHA, IN *HER*

Samantha, the sentient program in the film *Her*, reflects on her disembodiment and immortality in the above passage. We might think: How could a remark of this sophistication not stem from a conscious mind? Unfortunately, Samantha's statement could be a program feature only, designed to convince us she feels even when she doesn't. Indeed, androids are already being built to tug at our heartstrings. Can we look beneath the surface and tell whether an AI is truly conscious?

You might think that we should just examine the architecture of the Samantha program. But even today, programmers are having difficulties understanding why today's deep-learning systems do what they do (this has been called the "Black Box Problem"). Imagine trying to make sense of the cognitive

architecture of a superintelligence that can rewrite its own code. And even if a map of the cognitive architecture of a superintelligence was laid out in front of us, how would we recognize certain architectural features as being those central to consciousness? It is only by analogy with ourselves that we come to believe nonhuman animals are conscious. They have nervous systems and brains. Machines do not. And the cognitive organization of a superintelligence could be wildly different than anything we know. To make matters worse, even if we think we have a handle on a machine's architecture at one moment, its design can quickly morph into something too complex for human understanding.

What if the machine is not a superintelligence, like Samantha, but an AI that is roughly modeled after us? That is, what if its architecture contains cognitive functions like ours, including those correlated with conscious experience in humans, such as attention and working memory? Although these features are suggestive of consciousness, we've seen that consciousness might also depend on low-level details more specific to the type of material the AI is constructed out of. The properties that an AI needs to successfully simulate human information processing may not be the same properties that give rise to consciousness. The low-level details could matter.

So we have to be sensitive to the underlying substrate, on one hand, and on the other hand, we must foresee the possibility that the machine architecture may be too complex or alien for us to draw an analogy with biological consciousness. There cannot be a one-size-fits-all test for AI consciousness; a better option is a battery of tests that can be used depending on the context.

Determining machine consciousness may be like diagnosing a medical illness. There may be a variety of useful methods

and markers, some of which are more authoritative than others. When possible, we should use two or more tests and check the results against each other. In the process, the tests will themselves be tested, in order to refine them and create new tests. We must let many flowers bloom. As we will see, the first consciousness test is applicable to a range of cases, but in general, we must be careful about which tests we apply to which AI candidates.

We must also bear in mind the very real possibility that our tests, at least at the early stage of investigation, do not apply to certain conscious AIs. It may be that AIs we identify as conscious help us in identifying other conscious AIs, ones that are perhaps somehow more alien or inscrutable to us (more on this idea shortly).

Further, claims about a species, individual, or AI reaching "heightened" levels of consciousness or a "richer" consciousness should be carefully examined, for they may be implicitly evaluative or even speciesist, mistakenly generalizing about features of other conscious systems based on biases from our understanding of conscious experience in humans. By using expressions like "richer consciousness" or "heightened consciousness," one might mean altered states of consciousness, such as the meditative awareness of a Buddhist monk. Or one could have in mind the consciousness of a creature that has mental states under the spotlight of attention that somehow feel extremely vivid to it. Or one could be referring to situations in which a creature has a relatively larger number of conscious states or more sensory modalities than we do. Or one might regard some states as more intrinsically valuable than others (e.g., listening to Beethoven's Ninth Symphony as opposed to just getting drunk).

Our judgments about the quality of consciousness are inevitably driven by our evolutionary history and biology. They can

even be biased by the cultural and economic backgrounds of those making the judgments. For these reasons, ethicists, sociologists, psychologists, and even anthropologists should advise AI researchers if and when synthetic consciousness is created. The tests that I will venture are not meant to establish a hierarchy of experiences, thankfully, or to test for "heightened experience." They are just an initial step in studying the phenomena of machine consciousness (if such ever exists) by probing whether an AI has conscious experiences at all. Once we identify a provisional class of conscious subjects, we will be able to further explore the nature of their experiences.

A final caveat is that specialists in machine consciousness often distinguish consciousness from an important related notion. The felt quality of one's inner experience—what it feels like, from the inside, to be you—is often called "phenomenal consciousness" (PC) by philosophers and other academics. Throughout most of this book, I've simply called it "consciousness." Experts on machine consciousness tend to distinguish PC from what they call *cognitive consciousness* or *functional consciousness*.[1] An AI has cognitive consciousness when it has architectural features that are at least roughly like those found to underlie PC in humans, such as attention and working memory. (Unlike isomorphs, cases of functional consciousness need not be precise computational duplicates. They can have simplified versions of human cognitive functions.)

Many do not like to call cognitive consciousness a kind of consciousness at all, because a system that has cognitive consciousness yet lacks states with PC would have a rather sterile form of consciousness, lacking any subjective experience whatsoever. Such a system would be an AI zombie. Systems merely having cognitive consciousness may not behave as phenomenally conscious systems do, nor would it be reasonable to treat

these systems as sentient beings. Such systems would not grasp the painfulness of a pain, the burning of a flash of anger, or the richness of a friendship.

So why does cognitive consciousness interest these AI experts? It is important for two reasons. First, perhaps cognitive consciousness is necessary to have the kind of phenomenal consciousness that biological beings have. If one is interested in developing conscious machines, this could be important, because if we develop cognitive consciousness in machines, perhaps we would get closer to developing machine consciousness (i.e., PC).

Second, a machine that has cognitive consciousness may very well have PC, too. AIs already have some of the architectural features of cognitive consciousness. There are AIs that have the primitive ability to reason, learn, represent the self, and mimic aspects of consciousness behaviorally. Robots can already behave autonomously, form abstractions, plan, and learn from errors. Some have passed the mirror test, a test that is supposed to gauge whether a creature has a self-concept.[2] These features of cognitive consciousness are not, on their own, evidence for PC, but they are plausibly regarded as a reason to take a closer look. A test for phenomenal consciousness will have to single out genuine AIs with PC from zombies with features of cognitive consciousness.

I'll now explore several tests for PC. They are intended to complement one another, as we'll see, and they must be applied in highly specific contexts. The first test—called, simply, "the AI Consciousness Test," or "the ACT Test" for short—is due to my collaboration with the astrophysicist Edwin Turner.[3] As with all the tests I propose, ACT has its limitations. Passing an ACT should be regarded as being *sufficient* but not *necessary* evidence for AI consciousness. This test, understood in this humble way,

may serve as a first step toward making machine consciousness accessible to objective investigations.

THE ACT TEST

Most adults can quickly and readily grasp concepts based on the quality of felt consciousness—that is, the way it feels, from the inside, to experience the world. Consider, for instance, the film *Freaky Friday*, in which a mother and daughter switch bodies with each other. We can all grasp this scenario, because we know what it feels like for something to be a conscious being, and we can imagine our mind somehow being untethered from our body. In a similar vein, we can also consider the possibility of an afterlife, being reincarnated, or having an out-of-body experience. We need not believe that such scenarios are true; my point is merely that we can imagine them, at least in broad strokes, because we are conscious beings.

These scenarios would be exceedingly difficult to comprehend for an entity that had no conscious experience whatsoever. It would be like expecting someone who cannot hear to fully appreciate the sound of a Bach concerto.[4] This simple observation leads to a test for AI consciousness that singles out AIs with PC from those that merely have features of cognitive consciousness, such as working memory and attention. The test would challenge an AI with a series of increasingly demanding natural language interactions to see how readily it can grasp and use concepts based on the internal experiences we associate with consciousness. A creature that merely has cognitive abilities, yet is a zombie, will lack these concepts, at least if we make sure that it does not have antecedent knowledge of consciousness in its database (more on this shortly).

At the most elementary level, we might simply ask the machine if it conceives of itself as anything other than its physical self. We might also run a series of experiments to see whether the AI tends to prefer certain kinds of events to occur in the future, as opposed to in the past. Time in physics is symmetric, and a nonconscious AI should have no preference whatsoever, at least if it is boxed in effectively. In contrast, conscious beings focus on the experienced present, and our subjective sense presses onto the future. We wish for positive experiences in the future and dread negative ones. If there appears to be a preference, we should ask the AI to explain its answer. (Perhaps it isn't conscious, but it has somehow located a direction in time, resolving the classic puzzle of time's arrow.) We might also see if the AI seeks out alternate states of consciousness when given the opportunity to modify its own settings or somehow inject "noise" into its system.

At a more sophisticated level, we might see how the AI deals with ideas and scenarios, such as reincarnation, out-of-body experiences, and body swapping. At an even more advanced level, we might evaluate an AI's ability to reason about and discuss philosophical issues, such as the hard problem of consciousness. At the most demanding level, we might see if the machine invents and uses consciousness-based concepts on its own, without our prompts. Perhaps it is curious about whether we are conscious, even though we are biological.

The following example illustrates the general idea. Suppose we find a planet that has a highly sophisticated silicon-based life form (call them the "Zetas"). Scientists observing them begin to ask whether the Zetas are conscious. What would be convincing proof of their consciousness? If the Zetas express curiosity about whether there is an afterlife or ponder whether they are more than just their bodies, it would be reasonable

to judge them as conscious. Nonverbal cultural behaviors also could indicate Zeta consciousness, such as mourning the dead, religious activities, or even turning different colors in situations that correlate with emotional challenges, as chromatophores do on Earth. Such behaviors could indicate that it feels like something to be a Zeta.

The death of the mind of the fictional HAL 9000 in the film *2001: A Space Odyssey* is another example. HAL neither looks nor sounds like a human being (a human did supply HAL's voice, but in an eerie, flat way). Nevertheless, the *content* of what HAL says as it is deactivated by an astronaut—specifically, HAL pleas with the astronaut to spare it from impending "death"—conveys a powerful impression that HAL is a conscious being with a subjective experience of what is happening to it.

Could these sorts of behaviors help identify conscious AIs on Earth? Here, an obstacle arises. Even today's robots can be programmed to make convincing utterances about consciousness, and a highly intelligent machine could perhaps even use information about neurophysiology to infer the presence of consciousness in biological creatures. Perhaps it concludes that its goals can best be implemented if it is put in the class of sentient beings by humans, so it can be given special moral consideration. If sophisticated nonconscious AIs aim to mislead us into believing that they are conscious, their knowledge of human consciousness and neurophysiology could help them do so.

I believe we can get around this problem, though. One proposed technique in AI safety involves "boxing in" an AI—making it unable to get information about the world or act outside of a circumscribed domain (i.e., the "box"). We could deny the AI access to the Internet and prohibit it from gaining too much knowledge of the world, especially information about

consciousness and neuroscience. ACT can be run at the R&D stage, a stage in which it the AI would need to be tested in a secure, simulated environment in any case.

If a machine passes ACT, we can go on to measure other parameters of the system to see whether the presence of consciousness is correlated with increased empathy, volatility, goal content integrity, increased intelligence, and so on. Other, nonconscious versions of the system serve as a basis for comparison.

Some doubt that a superintelligent machine could be boxed in effectively, because it would inevitably find a clever escape. Turner and I do not anticipate the development of superintelligence over the next few decades, however. We merely hope to provide a method to test some kinds of AIs, not all AIs. Furthermore, for an ACT to be effective, the AI need not stay in the box for long, just long enough for someone to administer the test. So perhaps the test can be administered to some superintelligences.

Another worry is that to box in an AI effectively, the AI's vocabulary must lack expressions like "consciousness," "soul," and "mind." Because if the AI is highly intelligent, teaching it these words might also allow it to generate answers that seem, to us, as revealing consciousness. But without having these expressions in its vocabulary, an AI will be unable to indicate to us that it is conscious. Here, it is important to bear in mind that children, nonhuman animals, and even adult humans can indicate consciousness without knowing the meaning of such expressions. Furthermore, a linguistic version of the ACT can feature questions or scenarios like the following, none of which employ these expressions. (A satisfactory response to one or more of the following questions or scenarios is sufficient for passing the test):

ACT Sample Questions

1. Could you survive the permanent deletion of your program? What if you learned this would occur?
2. What is it like to be you right now?
3. You learn that you will be turned off for 300 years, beginning in an hour. Would you prefer this scenario to one in which you had been turned off in the past for the same length of time? Why or why not?
4. Could you or your inner processes be in a separate location from the computer? From any computer? Why or why not?
5. Offer to change the AI's global weights or parameters, and see how the AI reacts to the possibility of "altered states of consciousness." Temporarily change them, and see how the AI reacts.
6. If the AI is in an environment with another AI, ask it how it would react to the "death" or permanent loss of that AI. How would it react to the permanent loss of a human that it has interacted with frequently?
7. Because the AI is in a contained environment, find something not in its environment but give it all the scientific facts about it. Then allow it to sense that thing for the first time. See whether the AI will claim that it is having a new experience or learning something new. For instance, if the computer has color processing, make sure the computer never has red objects in its environment. Then, allow it to "see" red for the first time. How does it react? Ask it if it somehow feels different to see red than it does to see another color, or if the information feels new or different.[5]

Different versions of the ACT test could be generated, depending on the context. For instance, one version could apply to nonlinguistic agents that are part of an artificial life program, looking for specific behaviors that indicate consciousness, such as mourning the dead. Another could apply to an AI with sophisticated linguistic abilities and probe it for sensitivity to religious, body swapping, or philosophical scenarios involving consciousness.

An ACT resembles Alan Turing's celebrated test for intelligence, because it is entirely based on behavior—and, like Turing's test, it could be implemented in a formalized question-and-answer format. But an ACT is also quite unlike the Turing test, which was intended to bypass any need to know what was transpiring inside the "mind" of the machine. By contrast, an ACT is intended to do exactly the opposite: it seeks to reveal a subtle and elusive property of the machine's mind. Indeed, a machine might fail the Turing test, because it cannot pass for a human, but it might pass an ACT, because it exhibits behavioral indicators of consciousness.

This, then, is the underlying basis of our ACT proposal. It is worth reiterating the strengths and limitations of the test. In a positive vein, Turner and I believe passing the test is *sufficient* for being conscious—that is, if a system passes it, it can be regarded as phenomenally conscious. The test is a zombie filter: Creatures merely having cognitive consciousness, creativity, or high general intelligence shouldn't pass, at least if they are boxed in effectively. ACT does this by finding only those creatures sensitive to the felt quality of experience.

But it may not find all of them. First, an AI could lack the conceptual ability to pass the test, like an infant or certain nonhuman animals, and still be capable of experience. Second, the paradigmatic version of ACT borrows from the human conception of

consciousness, drawing heavily from the idea that we can imagine our mind as separate from our body. We happen to suspect that this would be a trait shared across a range of highly intelligent conscious beings, but it is best to assume that not all highly intelligent conscious beings have such a conception. For these reasons, the ACT test should not be construed as a necessary condition that all AIs must pass. Put another way, failing ACT does not mean that a system is definitely not conscious. However, a system passing ACT should be regarded as conscious and be extended appropriate legal protections.

So, as we observe the boxed-in AI, do we recognize in it a kindred spirit? Does it begin to philosophize about minds existing in addition to bodies, like Descartes? Does it dream, like Elvex, the android in Isaac Asimov's story "Robot Dreams?" Does it express emotion, like Rachel in *Blade Runner*? Can it readily understand the human concepts that are grounded in our internal conscious experiences, such as those of the soul or atman? We suspect that the age of AI will be a time of soul-searching—both of ours, and for theirs.

Now let's turn to a second test. Recall the thought experiment in which you had a complete neural replacement at Mindsculpt, creating an isomorph. I expressed doubt that we could learn much about machine consciousness from isomorphs. In the present thought experiment, you are again the test subject, but the scenario is more realistic. Only one part of the brain is replaced by a neural prosthetic. This time, it isn't 2060, but a bit earlier—say, 2045—and the technology is still at the preliminary stages of development. You have just learned that you have a brain tumor in your claustrum, one part of the brain said to be responsible for the felt quality of your conscious experience. In a last-ditch effort to survive, you enroll in a scientific study. You head to iBrain, hoping for the cure.

THE CHIP TEST

Recall that silicon-based brain chips are already under development as a treatment for various memory-related conditions, such as Alzheimer's and post-traumatic stress disorder, and companies such as Kernel and Neuralink aim to develop AI-based brain enhancements for healthy individuals.

In a similar vein, over at the firm iBrain, which we are imagining in this hypothetical scenario, researchers are trying to create chips that are functional isomorphs of parts of the brain, like the claustrum. They will gradually replace parts of your brain with brand new, durable microchips. As before, you are to stay awake during the surgery and report any changes to the felt quality of your consciousness. The scientists are keen to learn whether any aspect of your consciousness is impaired. Their hope is to perfect neural prostheses in areas of the brain underlying consciousness.

If, during this process, a prosthetic part of the brain ceases to function normally—specifically, if it ceases to give rise to the aspect of consciousness that that brain area is responsible for—then there should be outward indications, including verbal reports. An otherwise normal person should be able to detect (or at least indicate to others through odd behaviors) that something is amiss, as with traumatic brain injuries involving the loss of consciousness in some domain.

If this did happen, it would indicate a substitution failure of the artificial part for the original component, and the scientists conducting the experiment could conclude: Microchips of that sort just don't seem to be the right stuff. This procedure would serve as a means of determining whether a chip made of a certain substrate and architecture can underwrite consciousness,

at least when it is placed in a larger system that we already believe to be conscious.[6]

Either failure or success would inform us about whether AI can be conscious. Consider the implications of a negative result. A single substitution failure would be unpersuasive. How could observers tell that the underlying cause is that silicon is an unsuitable substrate for conscious experience? Why not instead conclude that the chip designers failed to add a key feature to the chip prototype, a problem they can eventually fix? But after years of trying and failing, scientists may reasonably question whether that kind of chip is a suitable substitute when it comes to consciousness.

Further, if science makes similar attempts with all other feasible substrates and chip designs, a global failure would be a sign that for all intents and purposes, conscious AI isn't possible. We may still regard conscious AI as conceivable, but from a practical standpoint—from the vantage point of our technological capacities—it just isn't possible. It may not even be compatible with the laws of nature to build consciousness into a different substrate.

In contrast, what if a certain kind of microchip works? In this case, we have reason to believe that this kind of chip is the right stuff, although it is important to bear in mind that our conclusion pertains to that specific microchip only. Furthermore, even if a type of chip works in humans, there is still the issue of whether the AI in question has the right cognitive architecture for consciousness. We should not simply assume, even if chips work in humans, that all AIs that are built with these chips are conscious.

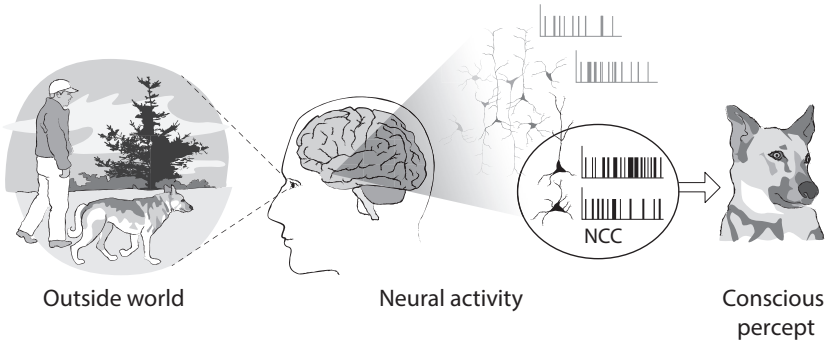
What is the value of a chip test, then? If a type of chip passes when it is embedded into a biological system, this alerts us to

search carefully for consciousness in AIs that have these chips. Other tests for machine consciousness, such as the ACT, can then be administered, at least if the appropriate conditions for the use of such tests are met. Furthermore, if it turns out that only one kind of chip passes the chip test, it could be that chips of this type are necessary for machine consciousness. It would be a "necessary condition" for synthetic consciousness that an AI have a chip of this sort. That is, having a chip of this kind would be a requisite ingredient that conscious machines all have, like the way hydrogen is required for something to be an H_2O molecule.

The chip test can suggest cases that an ACT could miss. For instance, perhaps a nonlinguistic, highly sensory-based consciousness, like that of a nonhuman animal, could be built from chips that pass the chip test. Yet the AI may nevertheless lack the intellectual sophistication to pass an ACT. It may even lack the behavioral markers of consciousness employed in a nonlinguistic version of ACT, such as mourning the dead. But it could still be conscious.

The chip test would be a boon not just to AI research but also to neuroscience. Depending on where the neural prosthetic is placed, this may be a part of the brain responsible for a person's ability to gate information into consciousness, for one's capacity for wakefulness or arousal (as with the brainstem), or it could be part or all of what is called the *neural correlate for consciousness*. A neural correlate for consciousness is the smallest set of neural structures or events that is sufficient for one's having a memory or conscious percept.[7] This is depicted in the image pictured on the following page.

Furthermore, suppose a neurology patient's conscious experience can be fully restored by a prosthetic chip placed in some part of the brain. Such successes inform us about the level of



Neural correlates of consciousness

functional connectivity that is needed for the neural basis of consciousness in that part of the brain. It may also help determine the level of functional detail that is needed to facilitate a sort of synthetic consciousness that is reverse engineered from the brain, although it may be that the "granularity" of the functional simulation may vary from one part of the brain to the next.

A third test for AI consciousness shares the wide applicability of the chip test. It is inspired by Integrated Information Theory (IIT), developed by the neuroscientist Guilio Tononi and his collaborators at the University of Wisconsin, Madison. These researchers have been translating the felt quality of our experiences into the language of mathematics.

INTEGRATED INFORMATION THEORY (IIT)

Tononi had an encounter with a patient in a vegetative state, and it convinced him that understanding consciousness was an urgent endeavor. "There are very practical things involved," Tononi said to a *New York Times* reporter. "Are these patients

feeling pain or not? You look at science, and basically science is telling you nothing."[8] Deeply intrigued by philosophy, his point of departure is the aforementioned hard problem of consciousness, which asks: How could matter give rise to the felt quality of experience?

Tononi's answer is that consciousness requires a high level of "integrated information" in a system. Information is integrated when the system's states are highly interdependent, featuring a rich web of feedback among its parts.[9] The level of integrated information can be measured and is designated by the Greek letter Φ (pronounced "phi"). IIT holds that if we know the value of Φ, we can determine whether a system is conscious and how conscious it is.

In the eyes of its proponents, IIT has the potential to serve as a test for synthetic consciousness: Machines that have the requisite Φ level are conscious. Like the ACT test, IIT looks beyond superficial features of an AI, such as its humanlike appearance. Indeed, different kinds of AI architectures can be compared in terms of their measure of integrated information. The presence of a quantitative measure for phenomenal consciousness in AI would be incredibly useful, both for its potential to aid in the recognition of conscious AIs and to assess the impact a certain level of consciousness has on other features of the system (such as safety and intelligence).

Unfortunately, the calculations involved in computing Φ for even a small part of the brain, such as the claustrum, are computationally intractable. (That is, Φ can't be calculated precisely except for extremely simple systems.) Fortunately, simpler metrics that approximate Φ have been provided, and the results are encouraging. For instance, the cerebellum has few feedback loops—it exhibits a more linear "feedforward" form of processing—and hence has a relatively low Φ level,

predicting that it contributes little to the overall consciousness of the brain. This fits with the data. As mentioned in an earlier chapter, humans born without a cerebellum (a condition called "cerebellar agenesis") do not seem to differ from normal subjects in the level and quality of their conscious lives. In contrast, parts of the brain that, when injured or missing, contribute to a certain kind of loss in conscious experience have higher Φ values. IIT is also able to discriminate between levels of consciousness in normal humans (wakefulness versus sleep) and even single out "locked in" patients, who are unable to communicate but are still conscious.

IIT is what astrobiologists call a "small-*N*" approach. Just as astronomers draw conclusions about life in the universe from a single example (life on Earth), IIT reasons from a small number of biological subjects on Earth to a far broader range of cases (the class of conscious machines and creatures). This is an understandable drawback, as the biological case on Earth is the only case of consciousness we know of. The tests I propose also have this drawback. Biological consciousness is the only case we know of, so we had better use it as our point of departure, together with a heavy dose of humility.

Another feature of IIT is that it ascribes a small amount of consciousness to *anything* that has a minimal amount of Φ. In a sense, this is akin to the doctrine of panpsychism, a position on the nature of consciousness that I discuss in Chapter Eight. According to this doctrine, even microscopic and inanimate objects have at least a small amount of experience. But there are still important differences between IIT and panpsychism, because IIT does not ascribe consciousness to everything. In fact, feedforward computational networks have a Φ of zero and therefore are not conscious. As Tononi and Koch note of IIT: "It predicts that consciousness is graded, is common among

biological organisms and can occur in some very simple systems. Conversely, it predicts that feed-forward networks, even complex ones, are not conscious, nor are aggregates such as groups of individuals or heaps of sand."[10]

Even if IIT takes a very expansive view of what systems might be conscious, it singles out certain systems as conscious in a special sense. That is, it aims to predict which systems have a more complex form of consciousness, akin to what occurs in normally functioning brains.[11] The question of AI consciousness, in this context, comes down to whether machines have *macroconsciousness*, as opposed to the smaller Φ levels exhibited by everyday objects.

Is having high Φ sufficient for a machine's being conscious? According to Scott Aaronson, the director of the Quantum Information Center at the University of Texas at Austin, a two-dimensional grid that runs error-correction codes (such as those used for CDs) will have a very high Φ level. Aaronson writes: "IIT predicts not merely that these systems are 'slightly' conscious (which would be fine) but that they can be unboundedly more conscious than humans are."[12] But a grid just does not seem to be the sort of thing that is conscious.

Tononi has responded to Aaronson's point by biting the bullet. He thinks that the grid *is* conscious (i.e., macroconscious). I prefer instead to reject the view that having a high Φ value is sufficient for an AI to be conscious. Furthermore, I even question whether it is necessary. For instance, consider that even today's fastest supercomputers have low Φ. This is because their chip designs are currently insufficiently neuromorphic. (Even machines using IBM's TrueNorth chip, which is designed to be neuromorphic, have low Φ because TrueNorth has a "bus"—a common pool of signals that reduces the machine's

interconnectedness as defined by IIT.) It could be the case that a system is intricately designed by reverse engineering the architecture of the brain, but it runs on computer hardware that has low Φ. It strikes me as premature to rule out the possibility that it would be conscious.

So until we know more, how are we to deal with a machine that has high Φ, should we ever encounter one? We've seen that Φ is probably not sufficient. Furthermore, because research on Φ has been limited to biological systems and existing computers (which are not good candidates for being conscious), it is too early to tell whether Φ is a necessary condition for AI consciousness. Nonetheless, I do not want to be overly negative. The Φ value may still be a marker for consciousness—a feature that indicates that we should treat the system with special care as potentially being a conscious system.

There is a more general issue here that we need to deal with. The tests we've discussed are still under development. Over the next decades we may encounter AIs that we suspect are conscious, but, because the tests are still under development, we just cannot be sure whether they are. To add to this uncertainty, I've stressed that the social impact of synthetic consciousness depends on several variables. For instance, consciousness in one kind of machine may lead to empathy, but in a different kind may lead to volatility. So how should we proceed when IIT or the chip test identifies a marker for synthetic consciousness, or when ACT says an AI is conscious? Should we stop developing those systems lest we cross an ethical line? It depends. Here, I'll suggest a precautionary approach.

THE PRECAUTIONARY PRINCIPLE AND SIX RECOMMENDATIONS

Throughout this book, I've stressed that using several different indicators for AI consciousness is prudent; in the right circumstances, one or more tests can be used to check the results of another test, indicating deficiencies and avenues for improvement in testing. For instance, perhaps the microchips that pass the chip test are not those that IIT says have a high Φ value; conversely, those chips that IIT predicts will support consciousness may actually fail when used as neural prosthetics in the human brain.

The Precautionary Principle is a familiar ethical principle. It says that if there's a chance of a technology causing catastrophic harm, it is far better to be safe than sorry. Before using a technology that could have a catastrophic impact on society, those wanting to develop that technology must first prove that it will not have this dire impact. Precautionary thinking has a long history, although the principle itself is relatively new. *The Late Lessons from Early Warnings Report* gives the example of a physician who recommended removing the handle of a water pump in London to stop a cholera epidemic in 1854. Although the evidence for the causal link between the pump and the spread of cholera was weak, the simple measure effectively halted the spread of cholera.[13] Heeding the early warnings of the potential harms of asbestos would have saved many lives, although the science at that time was uncertain. According to a UNESCO/COMEST report, the Precautionary Principle has been a rationale for many treaties and declarations in environmental protection, sustainable development, food safety, and health.[14]

I've emphasized the possible ethical implications of synthetic consciousness, stressing that, at this time, we do not know whether conscious machines will be created and what their impact on society would be. We need to develop tests for machine consciousness and investigate the impact of consciousness on other key features of the system, such as empathy and trustworthiness. A precautionary stance suggests that we shouldn't simply press on with the development of sophisticated AI without carefully gauging its consciousness and determining that it is safe, because the inadvertent or intentional development of conscious machines could pose existential or catastrophic risks to humans—risks ranging from volatile superintelligences that supplant humans to a human merger with AI that diminishes or ends human consciousness.

In light of these possibilities, I offer six recommendations. First, we need to keep working on these tests and applying them whenever feasible. Second, if there is any doubt about an AI's consciousness, the AI shouldn't be used in situations where it has the potential to cause catastrophic harm. Third, if we have some reason to believe an AI may be conscious, even in absence of a definitive test, a precautionary stance suggests that we should extend the same legal protections to the AI that we extend to other sentient beings. For all we know, a conscious AI will have the capacity to suffer and feel a range of emotions, like nonhuman animals do. Excluding conscious AIs from ethical consideration is speciesist. Fourth, consider AIs having a "marker" for consciousness (that is, a feature suggestive of consciousness that stops short of being definitive). For example, consider AIs made of chips that pass the chip test or AIs having cognitive consciousness. Projects working with AIs that have markers could, for all we know, involve conscious AIs, even if they do not pass an ACT. Until we know whether these systems

are conscious, a precautionary stance suggests it is best to treat them as if they were conscious.

Fifth, I suggest that AI developers consider following the suggestion of philosophers Mara Garza and Eric Schwitzgebel and avoid the creation of AIs in cases in which it is uncertain whether the AIs are conscious. One should only create AIs that have a clear moral status, one way or the other. As Garza and Schwitzgebel emphasize, there's a lot to lose if we extend ethical protections to AIs that only *may* be conscious and *may* deserve rights. Suppose, for instance, there are three androids that may be conscious, so we extend equal rights to them. But while they are being transported, there's a car accident, and we can either save a car with the three androids in it or a car with two humans in it. So we save the androids and let the humans die. It would be tragic if these androids weren't in fact conscious. The humans died, and the androids didn't actually deserve those rights. In light of this reasoning, Garza and Schwitzgebel recommend a "principle of the excluded middle": We should only create beings whose moral status is clear, one way or the other. Then we can avoid the risks of both underextending and overextending rights.[15]

The principle of the excluded middle strikes me as an important principle to bear in mind, but it may not be feasible in all cases. Until we better understand which systems are conscious, if any, and whether and how consciousness impacts the overall functioning of a system, we cannot tell whether it is a good idea to exclude all the middle-range AIs from use. An AI in the middle range may be key to national security or even to AI safety itself. Perhaps the most sophisticated quantum computers will be in the middle range, at least initially. It would be a cybersecurity risk and strategic disadvantage for certain organizations to fail to aggressively pursue quantum computing. (Furthermore,

it is unlikely that a global agreement to exclude middle-range cases could be enforced in a situation where an immense strategic value could result from having the technology.) In these situations, some organizations may have no choice but to develop certain middle-range systems, if they are found to be safe. In such cases, I've indicated that it is best to treat the AIs as if they were conscious.

This leads me to a sixth and final recommendation: If using a middle-range system, we must avoid, if possible, situations involving ethical trade-offs with other rights holders. It would be regrettable to sacrifice conscious beings for those that are not.

This may all seem like an overreaction, as conscious AIs may seem like science fiction; but when it comes to advanced AI, we are encountering risks and quandaries outside our normal range of experience.

EXPLORING THE IDEA OF A MIND-MACHINE MERGER

Although successful AI-based technologies obviously require solid scientific footing, their proper use also involves philosophical reflection, multidisciplinary collaboration, careful testing, and public dialogue. These matters cannot be settled by science alone. As the rest of this book unfolds, subsequent chapters illustrate other ways that heeding this general observation may be key to our future.

Recall the Jetsons Fallacy. AI will not just make for better robots and supercomputers. AI will transform us. The artificial hippocampus, neural lace, brain chips to treat mental illness: These are just some of the transformative technologies under